

Optimizing Creditworthiness Assessment: Leveraging PCA and ESVM for Improved Credit Score Analysis

Annie Chacko
Department of CSE
HITS, Padur,
Chennai.,India.
annievargh@gmail.com

D John Aravindhar
Department of CSE
HITS, Padur,
Chennai, India.
jaravindhar@hindustanuniv.ac.in

A.Antonidoss
Department of CSE and Business Systems
S.A Engineering College (Autonomous)
Chennai,India.
aro.antoni@gmail.com

Abstract -The paper proposes a novel approach to credit score analysis by combining Principal Component Analysis (PCA) with Ensemble Support Vector Machines (ESVM). This approach aims to address several limitations present in conventional credit score analysis techniques, including bias and fairness concerns, difficulties in handling unconventional credit data, lack of real-time data, inaccurate predictions, and unclear data. PCA is utilized to extract the most reliable and clear features from the dataset, which simplifies the classification process. By reducing the dimensionality of the data, PCA helps in improving the efficiency of the credit score analysis system. ESVM, a modified version of Support Vector Machines (SVM), is employed for classification tasks. SVM is known for its superiority in classification, and ESVM enhances its performance further by leveraging ensemble techniques. This combination enables accurate classification, thereby enhancing the accuracy of credit score predictions. To evaluate the proposed system, a credit scoring dataset is utilized in experiments. The results indicate that the introduced system outperforms conventional techniques in credit scoring analysis and offers better accuracy and reliability in assessing creditworthiness which providing financial institutions with better tools for risk management and decision-making.

Keywords-Credit score analysis, Support Vector Machine, Principal Component Analysis, Machine learning, Particle Swarm Optimization

I. INTRODUCTION

A credit score is very important in the financial industry to evaluate when a consumer is applying for a loan [1]. Traditional credit scoring methods include logistic regression, discriminant analysis, and linear regression. SVM is better than any other classifiers. The SVM method has been particularly effective in credit rating analysis. An ensemble model [2] is created using the optimized base classifiers that determine the classifier weights through particle swarm optimization. Data pre-processing, classifier

selection, and ensemble modeling are identified as useful steps to improve credit scoring model robustness and predictive performance. The data set demonstrates its consistency and accuracy, particularly for financial expenditure. In credit scoring with machine learning, accuracy and interpretability are key. The original KCDWU(K-means Clustering Distance Weighted Unsampling)model is adaptive but ignores the distances within and between classes, which may lead to inaccurate class feature identification. To address this, Automatic K-means clustering is improved using the Calinski-Harabasz index, resulting in improved clustering [3]. An innovative method involving weighted voting is proposed to improve the accuracy of outlier detection and a packing-based balanced sampling method is introduced to address imbalanced data problems [4]. One potential drawback of this solution is its high implementation cost, which may strain limited resources. Motivated by this, this paper employs Principal Component Analysis (PCA) and ensemble Support Vector Machine for the credit score analysis that solves this issue.

The paper organization of this research work is scheduled in section II as a literature review on Credit score analysis, drawbacks and challenges of the existing papers, and the research gap of this paper.

Section III explains the proposed methodology which includes different methods to secure high performance. The results and discussion section are depicted in section IV. Finally, this risk prediction paper is concluded in section V with future works.

II. RELATED WORKS

For credit scoring according to the ensemble classifiers and feature selection algorithm, Nalić et al. [6] proposed a novel hybrid data mining mechanism.

For assisting the decision-making procedure, a hybrid data mining mechanism with ensemble learning categorization algorithms and multiple feature selection was the objective of this work. The preprocessing of the datasets were done in the first phase. According to the output, five algorithms for feature selection were used. A novel voting technique named if-any was also proposed in this work. Four categorization algorithms were employed in the next phase. From the evaluation results, found that the proposed mechanism outperformed the existing single categorizer and other ensemble models. Various advanced credit scoring techniques were not explored in this work. For credit scoring, Zhang et al. [5] developed a novel hybrid ensemble mechanism that contains balanced sampling and outlier detection based on voting. The novel outlier detection based on voting was used to protect the data from noise. To forecast credit scoring, Plawiak et al. [7] established a hierarchical network of learners. The financial industry and banks used credit scoring (CS) as it was an important procedure in the management of risk. The CS was enhanced with various data mining (DM) and machine learning (ML) algorithms. A new technique known as the deep genetic hierarchical network of learners (DGHNL) was proposed in this article. From the evaluations done, it noted that the proposed technique attained a forecasting accuracy of 94.60%. Only a limited number of dataset was explored in this work. For credit scoring, Singh et al. [8] established an ensemble approach based on altered PSO clustering and multi-level classification. The accuracy of classification was affected by the unrelated features. Therefore, a new technique known as multi-level classification and cluster-based ensemble (MLCCE) which incorporates the characteristics of classification based on ensemble and feature selection was proposed in this article. For credit scoring, Abdoli et al. [9] employed a bagging supervised auto encoder classifier. One of the important processes adopted by the financial organizations was the automatic credit scoring. To solve this problem, a bagging supervised auto encoder classifier (BSAC) was proposed in this article.

Research Gap

To obtain a higher performance in the credit score analysis, this paper developed models for the process of credit score analysis.

Effective feature extraction:

Feature Extraction is the necessary step after the feature selection, so, Principal Component Analysis (PCA) is utilized.

Classification of features with higher performance:

After the feature extraction, classification is conducted by utilizing Ensemble Support Vector Machine (ESVM).

III. PROPOSED METHODOLOGY

The steps that are conducted in the classification of the credit score are demonstrated in Figure 1. In starting, the data like loan details, payment history, loan amount, etc that are required are gathered in the selected dataset, which is the credit score dataset. After collecting data, those data are pre-processed by using three pre-processing steps, such as data normalization, data reduction, and data cleaning. Then, the recursive feature selection is carried out for the selection of data from the pre-processed data. Principal Component Analysis (PCA) is deployed for the feature selection and the Ensemble Support Vector Machine is utilized for the classification of the credit scores.

A. Data Pre-processing

Data pre-processing is a basic step that includes gathering raw data and providing quality input for further processing. This work comprises three pre-processing steps.

Min-Max Normalization: Normalization is beneficial in the classification and also accelerates the training stage of the model. The following equation shows the min-max normalization:

$$N_q = \frac{N - N_{\text{minimum}}}{N_{\text{maximum}} - N_{\text{minimum}}} \quad (1)$$

In this, maximum and minimum are considered as 0 and 1.

Data Reduction: In this step, the size of the dataset will be reduced when keeping the significant features; this could be done by feature extraction.

Data Cleaning: In this data cleaning, removing misspellings and cleaning missing data in the dataset are included, and then it overcomes the data inconsistencies.

B. Recursive feature selection

In the feature selection process, we have introduced a recursive feature selection method for ranking the features based on their significance to mitigate feature redundancy and minimize computational expenses. Support vector machine-recursive feature elimination (SVM-RFE) plays vital role and it has been established as a potent technique for reducing dimensionality.

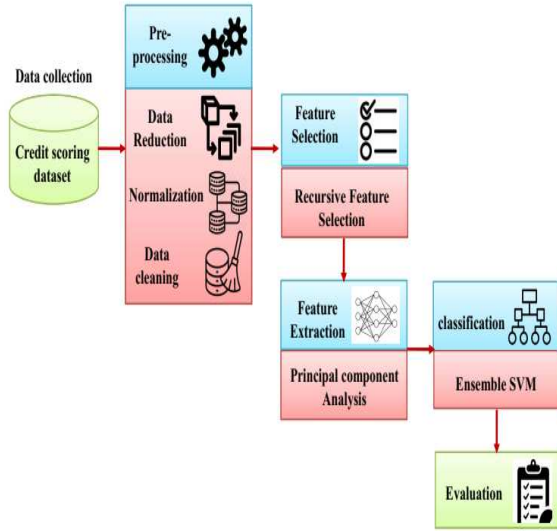


Fig 1: Proposed Architecture

C. Feature Extraction using Principal Component Analysis (PCA)

To extract the features, Principal Component Analysis (PCA) [10] was employed on the Xinjiang River. PCA involves a series of five primary operational steps, which can be outlined as follows: The initial data matrix is provided as:

$$Y = (y_{jk})_{o \times q} = \begin{bmatrix} y_{11} & \dots & y_{1q} \\ \vdots & \vdots & \vdots \\ y_{o1} & \dots & y_{oq} \end{bmatrix} \quad (2)$$

Within this matrix, each element denoted as y_{jk} represents the measured original data, where o corresponds to the monitoring station, and q signifies each specific quality parameter.

$$y_{jk}^* = (y_{jk} - \overline{y_k}) / t_k \quad (3)$$

This formula y_{jk}^* represents the standardized variable, $\overline{y_k}$ denotes average rate for the k^{th}

indicators, and t_k denotes the standard deviation of the k^{th} indicator. Compute the correlation coefficient matrix, denoted as S , using standardized data, and assess the relationships between the various indicators.

$$S = (s_{jk})_{q \times q} = \frac{1}{o-1} \sum_{u=1}^o y_{uj}^* \cdot y_{uk}^* \quad (j, k = 1, 2, \dots, q) \quad (4)$$

Using the correlation coefficient matrix S to find eigen values and eigenvectors to regulate the count of principal components. The matrix S has eigen values that are expressed as $\lambda_j (j = 1, 2, \dots, o)$ and eigen vectors as $v_j (v_j = v_{j1}, v_{j2}, \dots, v_{jo}) (j = 1, 2, \dots, o)$. λ represents variance, and higher variance indicates greater contribution to principal components. To ensure significant contribution, the cumulative rate of the first n components should be over 80%. This is expressed as $\sum_{k=1}^n \lambda_k / \sum_{k=1}^o \lambda_k \geq 0.80$. The principal component is denoted as:

$$G_j = v_{j1}y_1^* + v_{j2}y_2^* + \dots + v_{jo}y_o^* \quad (j = 1, 2, \dots, n), \quad (5)$$

The given equation y_j^* represents the standardized indicator variable. It is calculated as follows:

$$y_j^* = (y_j - \overline{y_j}) / t_j \quad (6)$$

The principal components acquired are combined by applying weights, then,

$$G = \frac{\lambda_1}{\lambda_1 + \lambda_2 + \dots + \lambda_o} G_1 + \frac{\lambda_2}{\lambda_1 + \lambda_2 + \dots + \lambda_o} G_2 + \dots + \frac{\lambda_o}{\lambda_1 + \lambda_2 + \dots + \lambda_o} G_o \quad (7)$$

D. Classification using Ensemble Support Vector Machine

One of the algorithms based on the kernel is the SVM [11]. It is an advanced technique that maximizes the margin by studying a discriminating hyperplane and for classification; it creates a better generalization capability. For two-class issues, the SVM is developed. The real examples m and related class labels are indicated as a set $\{(C_r, b_r), r = 1, 2, \dots, m\}$, here the r -sample's w -dimensional input features are denoted as $c_r \in I^w$, and the label for the negative and

positive class that contains the decision function $u(c) = d^G \theta(c) + y$ is $b_r \in \{+1, -1\}$. The following equation shows the main issue of soft-margin:

$$\min_{d,y} \frac{1}{2} \|d\|^2 + x \sum_{r=1}^m \tau_r \quad (8)$$

$$s.t. b_r (d^G \theta(c_r) + y) \geq 1 - \tau_r, \text{ with } \tau_r \geq 0, r=1,2,\dots,m \quad (9)$$

Here, the bias and weight vector are denoted as y and d , the nonlinear function used is $\theta(c)$, slack variables are represented as σ_r , and the penalty factor is X . Utilizing the dual issue and Lagrange multipliers technique, the aforementioned minimizing issue can be attained, as $\|d\|^2$ is convex. This process is given in the following equation:

$$\min_{\delta} \frac{1}{2} \sum_{r=1}^m \sum_{q=1}^m \delta_r \delta_q b_r b_q \theta^G(c_r) \theta(c_q) - \sum_{r=1}^m \delta_r \quad (10)$$

$$s.t. \sum_{r=1}^m \delta_r b_r = 0, \text{ with } 0 \leq \delta_r \leq X, r=1,2,\dots,m \quad (11)$$

Here, the coefficient of the Lagrange multiplier is δ_r . $d = \sum_{r=1}^m \delta_r b_r \theta(c_r)$ is provided by the solution of equation (2), and y is the expected bias, placing back to the linear decision function $u(c) = d^G \theta(c) + y$ and computing the inner product, the $P(c_r, c_q) = \theta(c_r)^G \theta(c_q)$ is defined as the kernel function; $\text{sgn}(\cdot)$ represents the soft-margin SVM's decision function.

$$u(c) = \text{sgn} \left(\sum_{r=1}^m \delta_r b_r P(c_r, c) + y \right) \quad (12)$$

IV. EXPERIMENTAL RESULTS

In financial institutions and banks, credit scoring is very important and beneficial. This paper developed an effective model that can solve issues and obtain good performance. In this section, the results that were conducted for the assessment of the proposed model are presented including the experimental setup, evaluation measures, dataset description, and performance analysis. For the comparison analysis, in this work, several methods are employed such as a deep genetic hierarchical network of learners (DGHNL) [7], Modified harmony search-based random forest (MHS-RF) [8], Multi-Level Classification and Cluster-based Ensemble (MLCCE)

[8], and Bagging Supervised Auto encoder classifier (BSAC) [11].

A. Experimental Setup: On the PC containing the Microsoft Windows 7 operating system and 32 GB of RAM with the 3.2 GHz Intel CORE i7 processor overall experiments of the developed model are conducted by applying Python Version 3.6.

B. Parameter Setting: The parameters that are employed in this work are presented in this section. Table 1 presents the parameters of each model as well as their values.

TABLE 1: PARAMETERS OF MODEL

Parameters	Values
Multiclass Method	One-vs-one
Kernel function	Quadratic
Box Constraint Level	1

The comparative study between all existing classifiers and the varieties of each is conducted in the present work in terms of the evaluation measures, where the setting parameters of the used classifiers are reported in the Table 1. In the existing models, Multiclass method is one –vs–one, Kernel function is quadratic and Box constraint level is 1.

C. Evaluation Measures:

Accuracy, precision, recall, specificity, f1-score, and Brier score measures are the evaluation measures employed in this paper based on the credit score analysis. By these measures, the proposed model's performance is determined, and these measures are represented in the following sections with their mathematical expression. In the below equations, TE_{PV} is specified as the true positive, TE_{NE} is indicated as true negative, FA_{PV} is denoted as the false positive, and FA_{NG} is indicated as the false negative.

Accuracy (AR): Accuracy is the most commonly used measure in credit score analysis and it is determined by the average of each true class. The following equation shows the computation of the accuracy measure.

$$AR = \frac{TE_{PV} + TE_{NG}}{TE_{PV} + FA_{PV} + TE_{NG} + FA_{NG}} \quad (13)$$

Precision (PS): The precision is defined as the total of positive predictions that divides the number of true positives. The below equation specifies the calculation of the precision measure:

$$PS = \frac{TE_{PV}}{(TE_{PV} + FA_{PV})} \quad (14)$$

Specificity (SP): This measure is the ratio of the true negatives in the total data and is also referred to as the true negative rate, the below equation specifies the computation of the measure.

$$SP = \frac{TE_{NG}}{TE_{NG} + FA_{PV}} \quad (15)$$

Recall (RL): Recall is defined as the ratio of the positive class that is predicted among the positive class and it is represented in the below equation:

$$RL = \frac{TE_{PV}}{TE_{PV} + FA_{NG}} \quad (16)$$

F1-score: The F1-score measure is the predictive efficiency of the model and this measure is computed through two measures such as recall and precision.

$$F1 = 2 * \frac{(PS * RL)}{PS + RL} \quad (17)$$

Brier Score (BS): The accuracy of the predicted probability and calibration of the prediction performance is measured by this score and is presented in the following equation.

$$BS = \frac{1}{M} \sum_{j=1}^M (q_j - x_j)^2 \quad (18)$$

Here, x_j is represented as the true label, the sample is indicated as j , the number of records is denoted as M , and the probability prediction is denoted as q_j .

D. Dataset Description

The dataset deployed for this analysis is the credit scoring dataset and this section presents the details about the dataset. This credit-scoring dataset [12] comprises various factors like credit usage ratio, payment history, employment status, number of credit accounts, and education level.

E. Performance analysis

The performance rate of the proposed model which is presented in this section includes several graphical representations based on the selected evaluation measures for the credit score analysis. Fig 2 represents the accuracy analysis of the proposed model and reaches a higher accuracy of 98.8%. Existing methods achieve low accuracy rates in credit scoring and are illustrated in Fig 3. i.e .97.5%, the existing methods reach a lower precision rate than the proposed model.

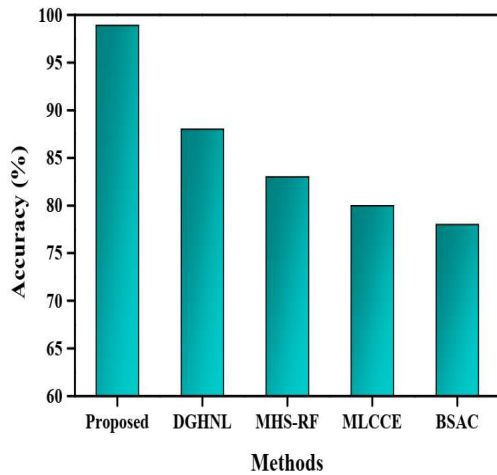


Fig 2: Accuracy analysis of the proposed model

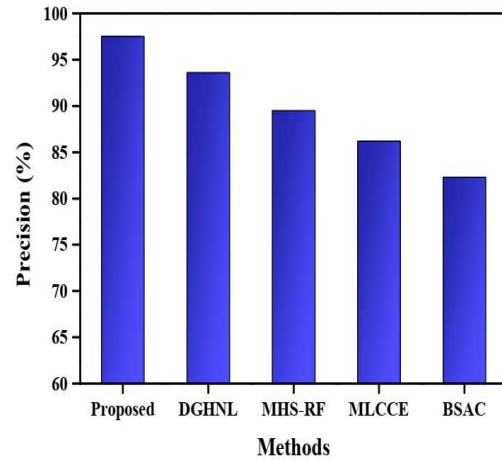


Fig 3: Precision analysis of the several methods

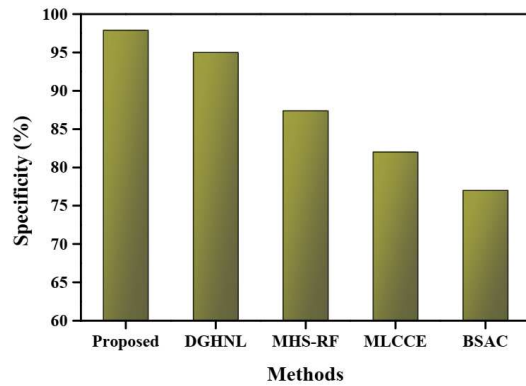


Fig 4: Specificity of the proposed model

Fig 4 depicts the graphical representation of the specificity of the proposed model. In the fig, the attained specificity of the proposed model has 97.9% in the comparison, the DGHNL, MHS-RF, MLCCE, and BSAC methods attain a lower performance on the specificity rates. Fig 5 demonstrates the performance based on the recall analysis of the proposed model and the recall rate of 96.1%. The F1-score analysis of the existing methods and proposed model is presented in Fig 6. The score for the

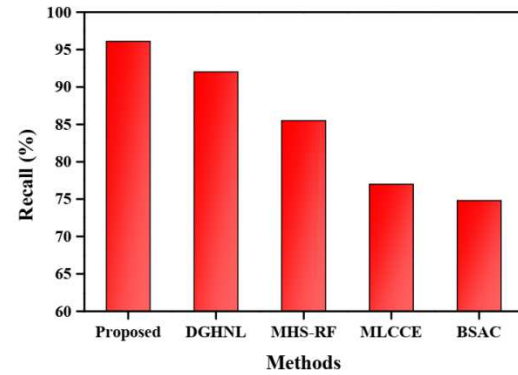


Fig 5: Effectiveness in terms of Recall analysis

proposed model is higher and the DGHNL, MHS-RF, MLCCE, and BSAC existing methods obtain a lower F1-score rate. The 97.1% is for the obtained F1-score rate of the developed model. Fig 7 depicts the Brier Score for the proposed model. The existing methods such as DGHNL, MHS-RF, MLCCE, and BSAC achieve a BS with higher rates. It can be seen, that the developed model's performance based on the BS is lower in these comparisons.

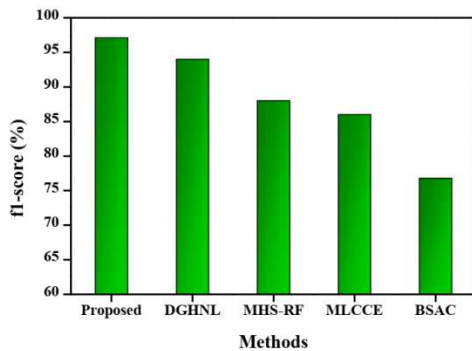


Fig 6: F1-score rate of the proposed model

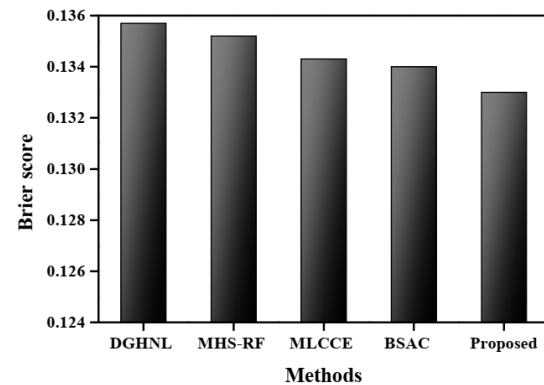


Fig 7: BS of the several methods

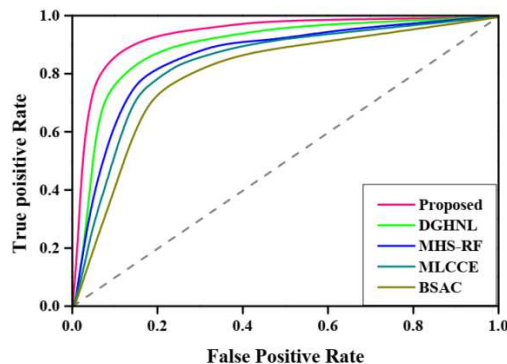


Fig 8: ROC analysis

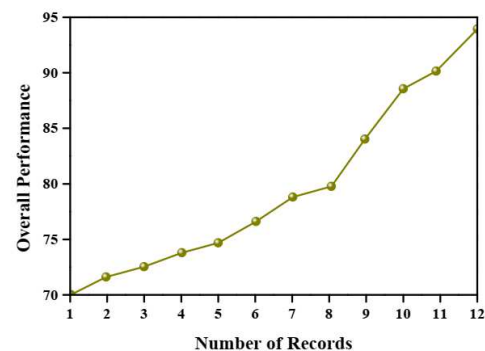


Fig9: Overall performance based on the features of the dataset

The Receiver Operating Characteristic (ROC) of the developed model is presented in Fig 8. In terms of credit scores, the proposed model has a 1.0 value in the Fig 8. The DGHNL, MHS-RF, MLCCE, and BSAC give a low performance in the ROC. The ROC Fig is illustrated as the True Positive Rate (TPR) and False Positive Rate (FPR). The overall performance of the developed model and other existing methods is demonstrated in Fig 9. The overall performance based on the records of the dataset is presented and is 98.9%.

V. CONCLUSION

For the credit score analysis in an effective manner, this paper uses the two techniques for achieving an accurate output. Principal Component Analysis (PCA) is utilized; this method extracts the features from the selected features. After this, the classification of these features is performed, for this, Ensemble Support Vector Machine (ESVM) is deployed. The evaluation of the proposed model is accomplished through evaluation measures including recall, accuracy, F1-score specificity, precision, and Brier Score (BS). According to these rates, the performance of the model is higher than the DGHNL, MHS-RF, MLCCE, and BSAC models. In future work, various optimized techniques will be combined with the proposed model to enhance the classification and the many credit-scoring datasets will be analyzed for the validation of the model.

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